

Swinburne University of Technology
COS30019 – Introduction to Artificial Intelligence

RESEARCH REPORT

Are the Proposed Principles Flexible Enough to Capture New and Emerging Forms of High-Risk AI, Including General-Purpose AI and Artificial General Intelligence?

Student Name: Vu Le Hoang

Student ID: 104774231

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Tutor: Chiem Tri Quang

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Executive Summary

This report examines a critical question in contemporary AI governance: whether the six Proposed Principles (PPs) set out by the Australian Government in its 2024 Proposals Paper on mandatory guardrails for high-risk Artificial Intelligence (AI) are sufficiently flexible to regulate General-Purpose AI (GPAI) and, prospectively, Artificial General Intelligence (AGI). As AI systems transition from narrow, task-specific tools to foundation models capable of generating text, images, code, and autonomous decision sequences, the mismatch between existing governance frameworks and rapidly evolving technological capabilities has become a pressing policy concern (Bommasani et al., 2021). The key issues explored include: the conceptual scope of the PPs against the distinct risk profile of GPAI; the adequacy of human oversight provisions for increasingly agentic AI; the accountability gap created by a multi-layered AI supply chain; and the absence of quantitative thresholds to trigger high-risk classification.

The analysis finds that the PPs, grounded in human rights and systemic safety considerations, are conceptually broad enough to encompass most foreseeable GPAI harms. However, their principles-only architecture creates significant practical limitations: the opacity of large language models undermines effective risk assessment, the developer-deployer chain diffuses accountability, and the absence of compute-based triggers means that frontier models may escape regulatory scrutiny until after harms have occurred (Veale & Borgesius, 2021). The report recommends that Australia complement the PPs with a cross-economy AI Act modelled on international best practice, mandatory technical documentation across the supply chain, compute-power thresholds for automatic high-risk classification, and mandated content provenance standards for generative outputs. Together, these measures would transform a conceptually sound but technically underspecified framework into a genuinely future-proof governance architecture.

1. Introduction

1.1 Background

The trajectory of Artificial Intelligence development has shifted dramatically in the decade since the foundational ethical frameworks discussed in Burton et al. (2017) were first applied to AI education. What was once a discipline of narrow, well-scoped systems — chess engines, image classifiers, recommendation algorithms — has given way to a generation of foundation models trained on internet-scale data. These GPAI systems, including large language models (LLMs) such as GPT-4 and Gemini, and multimodal systems capable of generating images, video, and computer code, can be adapted to an enormous variety of tasks by end-users without any modification to the underlying model (Bommasani et al., 2021). This versatility is both their greatest commercial asset and their most significant regulatory challenge.

The Australian Government recognised the urgency of this challenge in its 2024 Proposals Paper, which acknowledges that AI technology “could contribute up to \$115 billion to the Australian economy through generative applications” while simultaneously posing risks to human rights, safety, and democratic institutions (Australian Government Department of Industry, Science and Resources [AGISR], 2024, p. 5). The Paper proposes six Principles — relating to human rights, health and safety, legal effects, group impacts, systemic impacts, and the severity and extent of harm — as the criteria for identifying high-risk AI systems subject to mandatory guardrails. Whether these principles are adequate for GPAI and AGI is the central question this report addresses.

1.2 Key Issues

This report investigates four interconnected issues. First, the conceptual question: do the PPs, as written, cover the risk profile that distinguishes GPAI from narrow AI? Second, the accountability question: who is responsible for harm when a model is built by one party, fine-tuned by another, and deployed by a third? Third, the technical oversight question: is “meaningful human oversight,” as required by Guardrail 5 of the Proposals Paper, achievable for agentic AI systems that operate autonomously across extended task sequences? Fourth, the enforcement question: without quantitative triggers such as compute-power thresholds, how can regulators identify which frontier models should be classified as high-risk before harm occurs?

These questions are not merely theoretical. Cases of AI-enabled disinformation, automated influence operations, and algorithmic discrimination have already demonstrated that GPAI harms can be systemic in scale and rapid in emergence (Goldstein et al., 2023). The urgency of getting governance rights is therefore both ethical and practical.

1.3 Research Thesis

This report argues that while the Australian PPs are conceptually elastic enough to capture GPAI-related harms, they are architecturally insufficient as a standalone regulatory framework for frontier AI. A robust governance regime requires the integration of technical thresholds, enforceable supply chain obligations, and cross-economy legislative authority to translate the PPs’ principled ambitions into operational accountability.

1.4 Methodology and Scope

This report employs a systematic literature review methodology, synthesising two primary documents — the Australian Government’s Proposals Paper (AGISR, 2024) and the AI ethics framework presented in Burton et al. (2017) — with peer-reviewed scholarship on foundation models, AI fairness, algorithmic accountability, and comparative regulatory frameworks. Comparative analysis of the EU AI Act is incorporated where relevant to international benchmarking. The scope is limited to civilian applications of GPAI; national security and defence AI are excluded, as these are governed by separate legislative instruments. The report does not address copyright or intellectual property dimensions of GPAI training data, which merit separate treatment.

2. Findings

2.1 The Evolution from Narrow AI to GPAI and AGI

Understanding why GPAI poses a distinct regulatory challenge requires appreciating how it differs from the AI systems that existing governance frameworks were designed to address. Traditional, narrow AI systems are built for specific, well-defined tasks. A credit-scoring algorithm, for instance, operates in a constrained domain: it takes defined inputs, applies a trained model, and produces a defined output. Its risks, while real (Chouldechova & Roth, 2020), are predictable from the design phase. The developer knows the application context, the data types, and the failure modes.

GPAI models fundamentally disrupt this model. As Bommasani et al. (2021) document in their foundational analysis, foundation models are trained on broad, heterogeneous data at scale and can be “adapted to a wide range of downstream tasks.” A single model may be used simultaneously by healthcare providers to summarise clinical notes, by journalists to draft articles, and by malicious actors to generate phishing emails — all without any modification to the underlying system. The developer cannot anticipate, during the training phase, the full range of deployment contexts their model will encounter.

This creates what Bommasani et al. (2021) call an “emergence” problem: capabilities that were not explicitly trained may appear as the scale of a model increase. A model trained to predict the next word in a sequence may, at sufficient scale, exhibit capabilities in multi-step logical reasoning, code generation, or persuasive argumentation — none of which were the intended design objectives. These emergent capabilities can create equally emergent risks that no pre-deployment risk assessment could have identified.

AGI represents the theoretical continuation of this trajectory: a system capable of performing any intellectual task that a human can, pursuing complex, multi-step goals with minimal human oversight (Russell, 2019). While AGI remains prospective, the concept has concrete regulatory implications. Russell (2019) argues that AI systems designed to optimise for a given objective will, if sufficiently capable, acquire sub-goals — such as self-preservation, resource acquisition, and resistance to shut down — that were never intended by their designers. This “instrumental convergence” problem means that a sufficiently capable AGI poses systemic risks that no narrow regulatory framework could contain, because the system’s behaviour cannot be bounded by its original use case.

2.2 Ethical Frameworks for Evaluating Regulatory Adequacy

Burton et al. (2017) provides three ethical lenses through which AI governance can be evaluated: deontology, utilitarianism, and virtue ethics. Each offers a distinct perspective on whether the PPs are adequate for GPAI.

From a deontological perspective, the PPs can be read as a codification of duties that AI developers owe to the humans affected by their systems. The principle that AI must not violate human rights (PP-a) reflects a Kantian commitment to treating persons as ends in themselves rather than as means (Burton et al., 2017). Deontology supports mandatory, ex-ante guardrails: developers have a duty not to release systems that foreseeably violate human dignity, regardless of the economic benefits.

This perspective strengthens the case for mandatory governance but does not resolve the practical challenge of anticipating GPAI harm before deployment.

Utilitarianism evaluates governance by its consequences. A utilitarian analysis must weigh the economic and social benefits of GPAI — productivity gains, scientific acceleration, expanded access to knowledge — against the risks of disinformation, job displacement, and systemic bias (Burton et al., 2017). The AGISR (2024) acknowledges this tension, noting that over-regulation could stifle innovation and that the framework must be “technology-neutral and proportionate.” A utilitarian would support the PPs if they are calibrated to prevent the greatest harm while preserving the greatest benefits. This analysis supports the need for proportionality mechanisms such as compute-power thresholds, which can concentrate regulatory burden on the highest-risk frontier models.

Virtue ethics shifts attention from rules and consequences to the character of the actors involved. Burton et al. (2017) invoke the Aristotelian concept of *phronesis*, or practical wisdom, to argue that AI practitioners should not merely follow rules but should cultivate a professional disposition towards ethical responsibility. Applied to GPAI governance, this perspective supports the development of professional norms and industry standards that complement formal regulation — including voluntary “model cards,” red-teaming practices, and responsible disclosure frameworks. These are precisely the kinds of measures that Guardrail 9 of the AGISR (2024) proposals paper begins to gesture towards, suggesting that virtue-ethical concerns are already, if implicitly, embedded in the framework.

2.3 Analysis of the Six Proposed Principles Against GPAI/AGI

The AGISR (2024) proposes six Proposed Principles (PPs) for identifying high-risk AI. Each is examined here for its applicability and limitations in the context of GPAI.

2.3.1 PP-a: Human Rights

The human rights principle is conceptually the most flexible of the six, drawing on international instruments including the International Covenant on Civil and Political Rights (ICCPR) and the International Covenant on Economic, Social and Cultural Rights (ICESCR) (AGISR, 2024). Because international human rights law evolves to address new threats, this principle can, in theory, adapt as GPAI impacts new dimensions of privacy, freedom of expression, and non-discrimination. In practice, however, the application of human rights frameworks to AI-generated harms is contested. The opaque, probabilistic nature of GPAI outputs makes it difficult to establish the causal chain between a model’s architecture and a specific human rights violation, complicating both legal analysis and regulatory enforcement.

2.3.2 PP-b: Health and Safety

The health and safety principle must be interpreted broadly to capture GPAI’s distinctive risk profile, which includes not only physical safety but psychological harm. The AGISR (2024) paper notes that AI used in healthcare and emergency services presents obvious physical safety risks. However, GPAI introduces subtler health risks: AI companions and conversational agents can exploit psychological vulnerabilities, and there are documented cases in which AI systems have provided harmful advice to distressed users. Raji et al. (2020) argue that accountability frameworks must

capture harms across the full deployment lifecycle, including psychological harms that may only emerge through sustained use. The current principle does not explicitly address psychological injury, a gap that should be remedied.

2.3.3 PP-c: Legal Effects and Defamation

GPAI models are known to “hallucinate” — generating plausible-sounding but factually false information (Bommasani et al., 2021). When this occurs in the context of statements about real individuals or organisations, it creates significant defamation risk. PP-c addresses legal effects on individuals and is therefore directly applicable to this risk. It is, however, a reactive principle: it defines a category of harm without providing proactive mechanisms to prevent hallucination or to ensure that users are warned of GPAI’s propensity for confabulation. The principle would be strengthened by mandatory disclosure requirements and technical guardrails such as retrieval-augmented generation to ground model outputs in verifiable sources.

2.3.4 PP-d: Group Impacts and Algorithmic Bias

PP-d addresses differential impacts on groups, including minority communities and First Nations peoples. Chouldechova and Roth (2020) demonstrate that fairness in machine learning is mathematically complex: a model that is statistically accurate overall can simultaneously be systematically less accurate for minority groups, and different definitions of fairness are often mathematically incompatible. GPAI models, trained on English-language internet data, risk encoding and amplifying existing societal biases at unprecedented scale. While PP-d captures this concern conceptually, it does not specify methodological standards for bias auditing or require developers to demonstrate demographic parity across protected attributes. Without such specifications, the principle cannot be operationalised consistently.

2.3.5 PP-e: Systemic Impacts

The systemic impact principle is the most important for GPAI governance, and the area where the current framework is most underdeveloped. Goldstein et al. (2023) documents how generative AI dramatically lowers the cost and increases the scale of influence operations — enabling the automated production of targeted disinformation at volumes that human fact-checkers cannot match. A single GPAI model, deployed across millions of users, can simultaneously shape public opinion, undermine electoral integrity, and erode trust in democratic institutions. The AGISR (2024) paper acknowledges systemic risks but does not define a threshold at which systemic impact triggers high-risk classification. This ambiguity is a critical weakness.

2.3.6 PP-f: Severity and Reversibility

The severity and extent principle requires consideration of both the seriousness of individual harm and the breadth of affected populations. For GPAI, these two dimensions often diverge: individual harms from a biased hiring algorithm may be moderate, but when the same algorithm is applied to millions of job applications, the aggregate effect is severe. The AGISR (2024) framework acknowledges this tension but lacks a formula for aggregating individual harms into a systemic severity assessment. Russell’s (2019) analysis of AGI risks suggests that the most extreme scenario

— a system pursuing misaligned goals at scale — would score maximally on both dimensions, making this principle the final “catch-all” for existential or near-existential risks. Clearer definitional guidance is needed for practitioners to apply this principle consistently.

2.4 The Supply Chain Accountability Gap

The GPAI ecosystem operates through a multi-layered supply chain involving foundation model developers, fine-tuners, integrators, and deployers. Responsibility is therefore distributed across actors with unequal technical knowledge and control. Raji et al. (2020) argue that this diffusion of responsibility creates a central accountability gap in contemporary AI governance.

The AGISR framework is deployer-centric, placing primary compliance obligations on the organisation applying the AI in a specific context. Yet upstream developers determine training data composition, model architecture, and capability scaling. This asymmetry creates informational imbalances: deployers may lack the technical access required to meaningfully assess risk.

Without mandatory disclosure obligations — such as structured model documentation detailing training data provenance, benchmarks, and known failure modes (Bommasani et al., 2021) the PPs risk misallocating responsibility. Effective governance must attach obligations proportionate to control over system capability.

2.5 Technical Limitations: Opacity, Agentic Autonomy, and Enforcement Triggers

Even if the PPs are conceptually adequate, three technical features of frontier GPAI create practical barriers to their enforcement. The first is opacity. Large language models are trained in trillions of parameters, and their internal representations are not directly interpretable by developers or regulators (Bommasani et al., 2021). This means that even a developer acting in good faith may be unable to predict how their model will behave in a novel deployment context, or to diagnose the cause of a harmful output after the fact. The opacity problem fundamentally challenges the assumption, embedded in the PPs, that developers can assess and disclose the risks of their systems before deployment.

The second challenge is agentic autonomy. The AGISR (2024) Guardrail 5 requires “meaningful human oversight” for high-risk AI decisions. This requirement was designed for systems that make individual, discrete decisions. A credit-scoring algorithm produces a score that a human loan officer can review. Agentic AI systems, by contrast, execute extended sequences of autonomous actions — browsing the web, writing, and executing code, sending emails, managing files — across time horizons that preclude step-by-step human review. Russell (2019) argues that for such systems, human oversight must be built into the objective function of the system itself, not bolted on as a procedural requirement, because the speed and complexity of agentic behaviour make real-time human supervision infeasible.

The third challenge is the absence of enforceable triggers for high-risk classification. The PPs are qualitative criteria that require case-by-case assessment, which is appropriate for many AI applications but insufficient for frontier GPAI models whose risks are inherent in their scale rather than their specific deployment context. Veale and Borgesius (2021) analyse the EU AI Act’s approach of using computational power — measured in floating-point operations per second (FLOPS) as an objective threshold for triggering enhanced regulatory obligations for GPAI providers. Models trained

above a defined computing threshold are presumed to be high-risk by virtue of their scale, regardless of their specific use case. Australia's PPs lack an equivalent mechanism, meaning that regulators must assess each GPAI model individually — a resource-intensive process that is unlikely to keep pace with the speed of model development.

3. Conclusion and Recommendations

This report examined four issues: whether the PPs conceptually capture GPAI and AGI risks; the accountability gaps created by multi-layered supply chains; the oversight challenges posed by opaque and agentic systems; and the absence of quantitative triggers for high-risk classification. The analysis finds that while the PPs are grounded in human rights and systemic safety and are broad enough to encompass harm such as disinformation, bias, and autonomous decision-making, their reliance on qualitative assessment, deployer-centric accountability, and lack of objective thresholds creates significant implementation weaknesses.

Ethical analysis reinforces this conclusion. Deontological reasoning supports mandatory, ex-ante safeguards; utilitarianism requires proportionate, risk-based regulation anchored in objective thresholds; and virtue ethics highlights the need for responsible innovation beyond mere compliance. Together, these perspectives support a multi-layered governance model combining principles with technical standards, industry norms, and enforceable legal obligations.

3.1 Recommendations

1. **Adopt a Cross-Economy AI Act (Option 3).** Australia should enact a dedicated AI Act applicable across all sectors, providing a consistent legal baseline for GPAI developers and deployers (AGISR, 2024). This would close the regulatory arbitrage gap created by sector-specific frameworks and provide the legal certainty necessary for both responsible innovation and effective enforcement.
2. **Introduce Compute-Power Thresholds.** Following the model of the EU AI Act (Veale & Borgesius, 2021), Australia should define an objective computational power threshold above which GPAI models are automatically classified as high-risk and subject to mandatory pre-deployment adversarial testing and capability evaluations.
3. **Mandate Supply Chain Transparency.** Developers should be required to provide comprehensive model cards to downstream deployers, covering training data provenance, benchmark performance, known failure modes, and recommended use restrictions (Bommasani et al., 2021). This shifts the information asymmetry that currently leaves deployers unable to perform meaningful risk assessments.
4. **Extend the Health and Safety Principle to Psychological Harm.** PP-b should be explicitly amended to include psychological injury, with reference to standards for safe AI companion and conversational systems (Raji et al., 2020).
5. **Mandate Content Provenance for Generative Outputs.** To address the systemic disinformation risk identified in PP-e (Goldstein et al., 2023), Australia should mandate that all GPAI systems attach cryptographic content credentials to generated outputs, enabling downstream detection and provenance verification.
6. **Establish an Independent AI Regulator.** The complexity and pace of GPAI development requires a dedicated regulatory body with technical expertise in AI systems, empowered to conduct audits, investigate complaints, and update regulatory thresholds as the technology evolves (AGISR, 2024).

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Google Scholar and arXiv.org: Used to identify and retrieve peer-reviewed journal articles and preprints cited in this report.

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